

Lineage-Aware Root-Cause Analysis for Data Engineering Pipelines

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Summary: Data pipelines often fail silently, this means that every step reports success, but the data is wrong.

Spotting that something is wrong is well supported. Finding where it started, and why, is not.

This work tests whether giving a Language Model the pipeline's lineage helps it find the origin and cause of these issues.

Objectives / Work done / Results

The objective of this work is to test how much does lineage influence the ability to find the root cause of a data fault.

To test this we will need fault injection, so the true cause is always known and accuracy can be measured.

This dissertation will focus on the OMOP CDM data pipeline use case. It's a real ETL with several steps, where data quality matters.

To work with this type of data we already asked and received access to a Medical Database (MIMIC).

Next Steps / Challenges / Bottlenecks

Moving forward the objective is to start doing an in-depth State of the Art.

Focusing on:

- ETL, data pipeline and observability;
- Root-cause analysis, lineage-aware agents;
- OMOP CDM and healthcare data quality;

Only after this research can we effectively plan what we are going to do in the next months.

A challenge I expect is defining "the root cause" when a fault crosses several steps, or when more than one fault is involved.